

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTEEC302 - ELECTRICAL MACHINES-I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 What is Transformer?
 - a) Transformer is a device used to convert low alternating voltage to a high alternating voltage
 - b) Transformer is a device used to convert alternating current to direct current
 - c) Transformer is a device used to convert low alternating current to a high alternating current
 - d) Transformers are used only for low alternating voltageWhat is the function of a transformer?
 - a) Transformer is used to step down or up the AC voltages and currents
 - b) Transformer is used to step down or up the DC voltages and currents
 - c) Transformer converts DC to AC voltages
 - d) Transformer converts AC to DC voltages
- 3 Three units of single-phase transformers and a single three-phase transformer of the same rating will
 - a) Always be the same.
 - b) Never be the same.
 - c) Might be the same.
 - d) None of the above.
- 4 For a three-phase transformer, the line voltage is
 - a) Equal to the phase voltage
 - b) 1.73 times the phase voltage
 - c) 1/1.73 times the phase voltage
 - d) 3 times the phase voltage
- 5 Which of the following best describes electromechanical energy conversion?
 - a) Conversion of electrical energy to heat energy.
 - b) Conversion of mechanical energy to electrical energy only.
 - c) Conversion of electrical energy to mechanical energy only.
 - d) Conversion of electrical energy to mechanical energy and vice-versa.

- 6 What is the most common coupling field used in electromechanical energy conversion devices?
- Electric field
 - Magnetic field
 - Both electric and magnetic fields
 - None of the above
- 7 The main function of a commutator in a DC generator is to:
- Increase the generated voltage
 - Convert AC voltage to DC voltage
 - Reduce sparking at the brushes
 - Provide a path for field current
- 8 Which type of DC generator is characterized by having its field winding connected in series with the armature winding and the load?
- Shunt generator
 - Series generator
 - Compound generator
 - Separately excited generator
- 9 Which type of DC motor is known for its high starting torque and variable speed applications?
- Shunt motor
 - Series motor
 - Separately excited motor
 - Permanent magnet motor
- 10 What is back EMF in a DC motor?
- A voltage induced in the armature winding due to the rotation of the motor.
 - The voltage supplied to the motor.
 - The voltage across the field winding.
 - A voltage that opposes the applied voltage.
- 11 Universal motors are used in
- conveyor.
 - food mixer.
 - elevator.
 - refrigerator.
- 12 Which of the following materials is generally used for the construction of a variable reluctance stepper motor? <https://www.batuonline.com>
- Paramagnetic.
 - Diamagnetic.
 - Ferromagnetic.
 - Antiferromagnetic.

Q2. Solve the following.

- Derive the EMF Equation of single phase Transformer. 6
- Explain the parallel operation of Single phase transformer. What are the condition for parallel operation of transformer? 6

Q3. Solve the following.

- What are the cooling Methodology of Three phase Transformer. 6
- What is three winding transformer and draw equivalent circuit. 6

Q4. Solve Any Two of the following.

- A) Write the applications of singly excited and multiple excited magnetic system. 6
- B) State the advantages of electromechanical energy conversion principles. 6
- C) Draw and explain the construction and working principle of three phase transformer. 6

Q5. Solve Any Two of the following.

- A) Explain the various types of DC generator according to the method of excitation. Draw the equivalent circuit. 6
- B) Explain briefly armature reaction in DC generator. 6
- C) A 4 pole lap wound DC shunt generator has useful flux per pole 0.07wb. The armature winding consist of 220 turn, each turn having a resistance of 0.004 ohm. Calculate the terminal voltage when running at 900RPM if armature current is 50A. 6

Q6. Solve Any Two of the following.

- A) Compare between DC generator and DC motor. 6
- B) What are the various method of speed control of DC shunt motor. Explain in details speed control by varying flux per pole and armature resistance of this motor. 6
- C) Write a short note on: 1) Brushless DC motor. 2) Stepper Motor. 6

*** End ***